

Practice sheet: Inductors and Capacitors

This sheet is not marked for correctness, and AI tools are allowed. The device-free quiz is built from twins of these questions, so make sure you can do each one on your own.

- **Two-tier items** ask for an answer *and* a reason. Get both right; on the quiz each earns half the marks.
- **Predict first:** where asked, write your prediction before you calculate.
- **Confidence:** next to each answer, write S (sure) or U (unsure). This is not marked; it shows you what to revise.
- **Conventions:** unless a figure shows otherwise, each element's current is referenced in the direction of its voltage drop (passive sign convention), so $v = L di/dt$ and $i = C dv/dt$. An initial current or voltage that points against the reference is negative. Check every answer: put $t = 0$ into your result, and differentiate it back.

S1. An inductor in a dc circuit (two-tier). A 2 H inductor in a dc circuit that has been on for a long time carries a constant 3 A. What is the voltage across it?

- | | |
|----------|--------|
| a) 6 V | b) 0 V |
| c) 1.5 V | d) 3 V |

Reason:

- The voltage is $v = Li$.
- The voltage is $v = L di/dt$, and the current is not changing.
- The voltage is $v = i/L$.
- The voltage is the stored energy divided by the current, $w/i = Li/2$.

Follow-up: how much energy does the inductor store?

S2. Voltage from current. As in Nilsson Example 6.1. The current in a 0.5 H inductor is 0 for $t < 0$ and $i = 8te^{-10t}$ A for $t \geq 0$.

- When is the current largest, and what is it?
- Find $v(t)$ for $t > 0$.
- When does the voltage change polarity?
- Does the voltage change instantly at $t = 0$? Does the current?
- What is the largest energy the inductor stores?

S3. Current from voltage. As in Nilsson Example 6.2. A 2 H inductor carries $i(0) = -1$ A. For $t > 0$ the voltage across it is $v = 12e^{-3t}$ V.

- Find $i(t)$ for $t \geq 0$.
- When is the current zero?
- In what interval does the inductor deliver energy? Use the sign of $p = vi$.
- Find the stored energy at $t = 0$ and as $t \rightarrow \infty$. Equal values: is that a contradiction?

S4. A capacitor in a dc circuit (two-tier). A $5\ \mu\text{F}$ capacitor in a dc circuit that has been on for a long time has a constant $12\ \text{V}$ across it. What current flows in it?

- a) $60\ \mu\text{A}$
- b) $30\ \mu\text{A}$
- c) $0\ \text{A}$
- d) $2.4 \times 10^6\ \text{A}$

Reason:

- i) The current is the stored energy divided by the voltage, $w/v = Cv/2$.
- ii) The current is $i = Cv$.
- iii) The current is $i = v/C$.
- iv) The current is $i = C\ dv/dt$, and the voltage is not changing.

Follow-up: how much energy does the capacitor store?

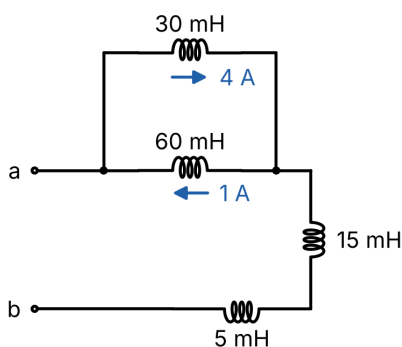
S5. Current from voltage. As in Nilsson Example 6.4. The voltage across a $4\ \mu\text{F}$ capacitor is 0 for $t < 0$ and $v = 20(1 - e^{-500t})\ \text{V}$ for $t \geq 0$.

- a) Find $i(t)$ for $t > 0$.
- b) Does the current change instantly at $t = 0$? Does the voltage?
- c) Find the energy stored as $t \rightarrow \infty$.
- d) When is the power into the capacitor largest, and what is it?

S6. A current pulse. As in Nilsson Example 6.5. A $0.5\ \mu\text{F}$ capacitor has $v(0) = 5\ \text{V}$. A current of $2\ \text{mA}$ flows into its $+$ terminal from $t = 0$ to $t = 5\ \text{ms}$, and zero afterwards.

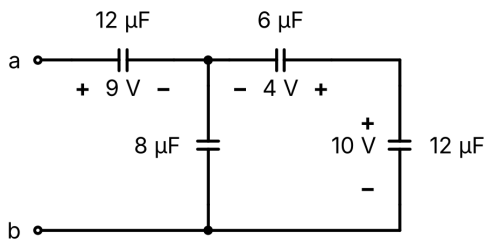
- a) Find $v(t)$ for $0 \leq t \leq 5\ \text{ms}$.
- b) What is v for $t > 5\ \text{ms}$? Why does it not return to $5\ \text{V}$?
- c) How much energy does the pulse deliver to the capacitor?
- d) *Predict first:* repeat (a)–(c) with a $-2\ \text{mA}$ pulse. When does v pass through zero?

S7. Equivalent inductance with initial currents. As in Nilsson Example 6.6.



- a) Find L_{eq} between a and b.
- b) Find the initial current in L_{eq} , entering at a.
- c) The $15\ \text{mH}$ and $5\ \text{mH}$ carry the current found in (b). Find the energy stored in the four inductors and in L_{eq} . Where is the difference?

S8. Equivalent capacitance with initial voltages. As in Nilsson Example 6.7.

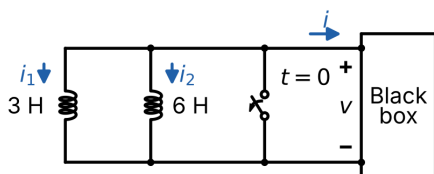


- Find C_{eq} between a and b.
- Find the initial voltage v_{ab} across C_{eq} .
- What is the initial voltage across the $8 \mu F$, + at the top? Find the energy stored in the four capacitors and in C_{eq} .

S9. Parallel capacitors. As in booklet Problem 6.9. A $0.4 \mu F$ and a $1.6 \mu F$ capacitor are in parallel. For $t \geq 0$ the voltage across them is $v = 50e^{-200t} + 30$ V. i_1 is the current in the $0.4 \mu F$ and i_2 the current in the $1.6 \mu F$, each referenced into the + terminal.

- Find C_{eq} .
- Find $i_1(t)$ and $i_2(t)$.
- Check that $i_1 + i_2 = C_{eq} dv/dt$. How do the two currents compare with the two capacitances?
- How much energy do the capacitors release between $t = 0$ and $t \rightarrow \infty$?

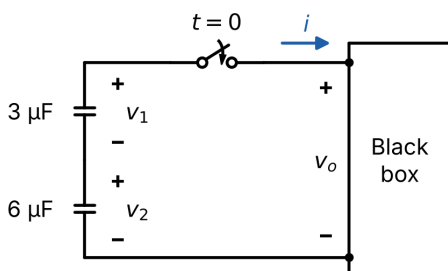
S10. Inductors and a black box. As in booklet Problem 6.8.



Just before the switch opens, $i_1(0) = -9$ A and $i_2(0) = 3$ A. For $t > 0$, $v = 36e^{-3t}$ V.

- Find L_{eq} and $i(0^+)$, the box current just after the switch opens.
- Find $i(t)$, $i_1(t)$ and $i_2(t)$ for $t \geq 0$.
- Find the initial energy stored in the inductors.
- Find the energy delivered to the black box and the energy trapped in the inductors.

S11. Capacitors and a black box. As in booklet Problem 6.10.



At $t = 0$, $v_1(0) = 20$ V and $v_2(0) = -8$ V. For $t > 0$, $i = 1.2e^{-50t}$ mA.

- Find C_{eq} and $v_o(0)$.
- Find $v_1(t)$, $v_2(t)$ and $v_o(t)$ for $t \geq 0$.
- Find the initial energy stored in the capacitors.
- Find the energy delivered to the black box and the energy trapped in the capacitors.

S12. Spot the error. The worked solution below contains the kind of direction slip AI chat tools often make. A 4 H inductor has its current i referenced in the direction of its voltage drop v . At $t = 0$ the current is 2 A, flowing against the reference arrow. For $t > 0$, $v = 24e^{-2t}$ V.

1. $i(t) = (1/L) \int v \, dt + i(0)$, integrating from 0 to t .
 2. $i(0) = 2$ A.
 3. The integral of $24e^{-2t}$ from 0 to t is $12(1 - e^{-2t})$.
 4. $i(t) = (12/4)(1 - e^{-2t}) + 2 = 5 - 3e^{-2t}$ A.
 5. Check: $i(0) = 2$ A and $4 \, di/dt = 24e^{-2t}$ V, so the answer is right.
- a) Which is the first wrong line?
 - b) Correct it and find $i(t)$.
 - c) Line 5 calls itself a check. Why did it not catch the error?
 - d) When does the current reverse direction?

S13. Spot the error in an energy balance. A 2 H and an 8 H inductor are in parallel, with $i_1(0) = 6$ A and $i_2(0) = -1$ A, both referenced downward; until $t = 0$ a switch across their terminals carries the total current. At $t = 0$ the switch opens and the current flows into a black box, which draws current until the total current is zero.

1. $L_{eq} = (2)(8)/(2 + 8) = 1.6$ H.
 2. The initial stored energy is $(1/2)(2)(6)^2 + (1/2)(8)(1)^2 = 36 + 4 = 40$ J.
 3. The box draws current until the inductors are empty, so it receives all the stored energy.
 4. The energy delivered to the box is 40 J.
- a) Which is the first wrong line?
 - b) Find the energy delivered to the box. (Use L_{eq} and the total initial current.)
 - c) Find i_1 and i_2 after a long time, and the energy trapped.

S14. A supercapacitor bank. A supercapacitor cell is rated 3000 F at 2.7 V.

- a) How much energy does one cell store at its rated voltage?
- b) Six cells are connected in series for a 16.2 V bank. Find C_{eq} and the energy the bank stores at 16.2 V. Compare with six times the energy of one cell.
- c) For how long could the bank supply 200 W if all of that energy could be used?
- d) *Predict first:* two such banks are put in parallel. What happens to C_{eq} and to the stored energy?

S15. Capacitive touch screens. As in Nilsson's Practical Perspective for this chapter. Each electrode of a touch screen has a parasitic capacitance $C_p = 20$ pF to ground. A finger adds $C_t = 5$ pF at the point of touch.

- a) What capacitance does the controller see at the touched electrode? By what percentage did it change?
- b) A self-capacitance screen is touched at (X1, Y2) and (X3, Y0) at the same time. Which four grid points look touched? Which two are ghosts?
- c) Why does a mutual-capacitance screen not produce ghost points?
- d) The controller charges an electrode with a constant 1 μ A. How long does the voltage take to reach 1 V when untouched, and when touched?

S16. Check an answer without solving again. A 0.25 H inductor carries $i(0) = 3$ A. For $t > 0$, $v = 10e^{-20t}$ V. Two classmates report $i(t)$ for $t \geq 0$. Student 1: $i = 5 - 2e^{-20t}$ A. Student 2: $i = 1 + 2e^{-20t}$ A. Use $i(0)$ and $v = L \, di/dt$ to decide which student is right.